

Estimating the Longest Increasing Sequence in Polylogarithmic Time

Rigorous Analysis, Definitions, and Proofs

Based on the paper by M. Saks and C. Seshadhri

Advanced Seminar Presentation (2-3 Hours)

Outline

- 1 Introduction and Formal Notation
- 2 Algorithmic Primitives: Splitters and Grids
- 3 The Basic Approximation Algorithm
- 4 Error Analysis: Primary and Secondary Error
- 5 The Improved Algorithm: Breaking the Logarithmic Barrier

The Problem and Results

- **Given:** An array represented as a function $f : [n] \rightarrow \mathbb{R}$.
- **Length of LIS:** lis_f . **Distance to Monotonicity:** $\text{loss}_f = n - \text{lis}_f$.
- **Theorem 1.1:** There is a randomized algorithm computing est such that $|\text{est} - \text{lis}_f| \leq \delta n$ with probability $\geq 3/4$ in time $(1/\delta)^{O(1/\delta)}(\log n)^c$.
- **Theorem 1.2:** There is a $(1 + \tau)$ -approximation to the distance to monotonicity running in time $(1/(\epsilon_f \tau))^{O(1/\tau)}(\log n)^c$.

This presentation focuses on the rigorous mathematical architecture proving these bounds.

Points and Partial Orders

Definition (Points and the Set $\mathcal{F}$)

We map the array f to points in $\mathbb{N}^2$. Let $F(x) = \langle x, f(x) \rangle$. We define $\mathcal{F} = \{F(x) : x \in (0, n]\}$.

Definition (Relations between Points)

For points P, Q :

- $P \preceq Q$: $x(P) \leq x(Q)$ and $y(P) \leq y(Q)$.
- $P \prec Q$: $x(P) < x(Q)$ and $y(P) \leq y(Q)$.
- $P \searrow Q$: $x(P) < x(Q)$ and $y(P) > y(Q)$ (a **violation**).

For $\mathcal{F}$ -points, $P \sim Q$ denotes they are comparable (either $P \prec Q$ or $Q \prec P$). $P \not\sim Q$ denotes a violation.

Regions Relative to a Point

For a given point P , we partition the plane $\mathbb{N}^2$ into four sets:

- $P^{SW} = \{Q \mid Q \preceq P\}$ (Southwest)
- $P^{NE} = \{Q \mid P \prec Q\}$ (Northeast)
- $P^{NW} = \{Q \mid x(Q) \leq x(P) \text{ and } y(Q) > y(P)\}$ (Northwest)
- $P^{SE} = \{Q \mid P \searrow Q\}$ (Southeast)

Note: If $Q \in P^{NW} \cup P^{SE}$, then Q and P are in violation ($Q \not\prec P$).

Boxes and Box Chains

Definition (Box)

A box $\mathcal{B} = X(\mathcal{B}) \times Y(\mathcal{B})$ where $X(\mathcal{B})$ is an index interval $(x_L, x_R]$ and $Y(\mathcal{B})$ is a value interval $[y_B, y_T]$. The *width* of $\mathcal{B}$ is $|X(\mathcal{B})| = x_R - x_L$.

Definition (Box Chain)

Boxes $\mathcal{B}_1 \triangleleft \mathcal{B}_2$ if the top-right of $\mathcal{B}_1$ is the bottom-left of $\mathcal{B}_2$. A sequence $\vec{\mathcal{T}} = (\mathcal{T}_1, \dots, \mathcal{T}_k)$ satisfying $\mathcal{T}_1 \triangleleft \dots \triangleleft \mathcal{T}_k$ is a **box chain**.

Any increasing sequence of points $P_0 \prec P_1 \prec \dots \prec P_k$ uniquely defines a box chain spanning from P_0 to P_k via $\text{Box}(P_{i-1}, P_i)$.

Strips and Local LIS

Definition (Strips)

A $\mathcal{B}$ -strip is a subbox $\mathcal{S} \subseteq \mathcal{B}$ such that $Y(\mathcal{S}) = Y(\mathcal{B})$. It represents a vertical slice of $\mathcal{B}$. If $\mathcal{T} \subset \mathcal{B}$, $\mathcal{T}|\mathcal{B}$ denotes the $\mathcal{B}$ -strip spanning the index interval of $\mathcal{T}$.

Definition (Local lis and loss)

For any box $\mathcal{B}$:

- $\text{lis}(\mathcal{B})$ is the size of the longest $(F, \mathcal{B})$ -increasing sequence.
- $\text{loss}(\mathcal{B}) = |\mathcal{B} \cap \mathcal{F}| - \text{lis}(\mathcal{B})$.

Splitters: Balance and Safeness

To simulate dividing and conquering the space, we search for indices to split a box $\mathcal{T} \subseteq \mathcal{B}$.

Definition (ρ -Balanced)

An index $s \in X(\mathcal{T})$ is ρ -balanced in $\mathcal{T}$ if $s - x_L(\mathcal{T}) \geq \rho \cdot w(\mathcal{T})$ and $x_R(\mathcal{T}) - s \geq \rho \cdot w(\mathcal{T})$.

Definition (Safeness Parameters)

Let $\mu \in (0, 1)$, $L > 0$. Let $viol(s, \mathcal{S})$ be the number of points in $\mathcal{S} \cap \mathcal{F}$ violating $F(s)$.

- We define $Z(s, \mathcal{S}) = viol(s, \mathcal{S}) - \mu|\mathcal{F} \cap \mathcal{S}|$.
- A strip $\mathcal{S}$ is (μ, L) -accepting if $Z(s, \mathcal{S}) \leq L$.

Adequate Splitters

Definition $((\mu, L, \rho)$ -Adequate Splitter)

Let $\mathcal{T} \subseteq \mathcal{B}$ be boxes. A splitter $s \in X(\mathcal{T})$ is (μ, L, ρ) -adequate if:

- 1 $F(s) \in \mathcal{T}$.
- 2 s is ρ -balanced in $\mathcal{T}$.
- 3 s is (μ, L) -safe for the strip $\mathcal{T}|\mathcal{B}$ (meaning every adjacent strip $\mathcal{S} \subseteq \mathcal{T}|\mathcal{B}$ is (μ, L) -accepting).

FindSplitter Procedure

Randomly samples $O(\log^2 n / \rho)$ balanced points. For each, it estimates $Z(s, \mathcal{S})$ via random sampling *approxZ*. By Hoeffding bounds, if an adequate splitter exists with frequency $\geq \rho$, it finds a valid splitter w.h.p.

The Dichotomy Lemma: Setup

The most critical structural lemma of the paper. If a box lacks good splitters, the LIS must lose a significant fraction of points.

- Let $\mathcal{R}$ be a box with a strip decomposition $\vec{S}$. Let $s(x)$ be the width of the strip containing x .
- An index x is *safe* if it is a $(\mu, s(x))$ -safe splitter for $\mathcal{R}$. Otherwise, *unsafe*.
- Let $\vec{\mathcal{D}}$ be an arbitrary box-chain compatible with $\vec{S}$.
- Let S be the safe indices in $\vec{\mathcal{D}}^\cup$, and U be the unsafe indices in $\vec{\mathcal{D}}^\cup$.
- Let $\chi = |\mathcal{R} \cap \mathcal{F}|$. χ^{in} points inside $\vec{\mathcal{D}}^\cup$, χ^{out} points outside.

The Dichotomy Lemma (Lemma 4.6)

Lemma 4.6

Under the defined setup, the number of points outside the chain satisfies:

$$\chi^{out} \geq \frac{\mu}{1 - \mu} |U|$$

Consequently, the number of points inside satisfies:

$$\chi^{in} \leq (1 - \mu)\chi + \mu|S|$$

Significance: If there are few safe points ($|S|$ is small), then χ^{in} is bounded strictly away from χ . Any increasing sequence MUST miss a fraction of points proportional to μ .

Proof of the Dichotomy Lemma (1/3)

Proof.

For each unsafe index $x \in U$, there exists an adjacent strip $\mathcal{W}(x)$ that is $(\mu, s(x))$ -rejecting. This means $\text{viol}(x, \mathcal{W}(x)) \geq \mu|\mathcal{F} \cap \mathcal{W}(x)| + s(x)$.

Let $\mathcal{V}(x)$ be the set of violations with $F(x)$ contained in $\mathcal{W}(x)$ but *not* in $\vec{\mathcal{D}}^U$.

Claim: $|\mathcal{V}(x)| \geq \frac{\mu}{1-\mu} |\vec{\mathcal{D}}^U \cap \mathcal{W}(x) \cap \mathcal{F}|$.

Proof of Claim: Any violation with x contained *inside* $\vec{\mathcal{D}}^U$ must reside in the single box $\vec{\mathcal{D}}[x]$. The number of such points is at most $w(\vec{\mathcal{D}}[x]) \leq s(x)$. □

Proof of the Dichotomy Lemma (2/3)

Proof.

Therefore, subtracting the internal violations, the external violations satisfy:

$$|\mathcal{V}(x)| \geq \mu|\mathcal{F} \cap \mathcal{W}(x)| + s(x) - s(x) = \mu|\mathcal{F} \cap \mathcal{W}(x)|$$

Since $\mathcal{V}(x)$ is disjoint from $\vec{\mathcal{D}}^U$, we have:

$$|\vec{\mathcal{D}}^U \cap \mathcal{W}(x) \cap \mathcal{F}| \leq (1 - \mu)|\mathcal{F} \cap \mathcal{W}(x)|$$

Rearranging gives $|\mathcal{V}(x)| \geq \frac{\mu}{1-\mu} |\vec{\mathcal{D}}^U \cap \mathcal{W}(x) \cap \mathcal{F}|$. □(Claim) □

Proof of the Dichotomy Lemma (3/3)

Proof.

We greedily select a subset $L' \subseteq L$ (unsafe indices facing left) and $R' \subseteq R$ (unsafe facing right) such that the strips $\mathcal{W}(z)$ are pairwise disjoint, yet they cover all $F(x)$ for $x \in U$.

The external violations $\mathcal{V}(x)$ lie in the Northwest or Southeast regions, completely disjoint from the box chain $\vec{\mathcal{D}}$.

$$\begin{aligned}\chi^{out} &\geq \left| \bigcup_{x \in L' \cup R'} \mathcal{V}(x) \right| = \sum_{x \in L' \cup R'} |\mathcal{V}(x)| \\ &\geq \sum_{x \in L' \cup R'} \frac{\mu}{1-\mu} |\vec{\mathcal{D}}^U \cap \mathcal{W}(x) \cap \mathcal{F}| \geq \frac{\mu}{1-\mu} |U|\end{aligned}$$

Since $\chi^{in} = |U| + |S|$, substituting $\chi^{in} = \chi - \chi^{out}$ yields $\chi^{in} \leq (1-\mu)\chi + \mu|S|$. □

Value Nets and Grids

To compute boosting paths, we partition terminal boxes.

Definition (Value Net)

An α -value net for $\mathcal{B}$ is a subset $V \subset Y(\mathcal{B})$ such that any subinterval J containing at least $\alpha \cdot w(\mathcal{B})$ points contains at least one element of V .

Definition (α -fine Grid)

A grid Γ is α -fine for $\mathcal{B}$ if:

- The columns partition $X(\mathcal{B})$ into strips of width $\leq \alpha \cdot w(\mathcal{B})$.
- The rows form an $(\alpha/|X(\Gamma)|)$ -value net.

Constructed via random sampling in procedure **BuildGrid**.

Grid Approximation Lemma

Lemma 4.9

If Γ is an α -fine grid for $\mathcal{B}$, and $\mathcal{L}$ is an increasing sequence in $\mathcal{B}$, there exists a $D(\Gamma)$ -chain $\vec{\mathcal{D}}$ such that:

$$|\mathcal{L} \setminus \vec{\mathcal{D}}^U| \leq \alpha \cdot w(\mathcal{B})$$

Proof Sketch.

Let $x_1 < \dots < x_r$ be the columns of Γ . Let Q_i be the largest point of $\mathcal{L}$ with index $\leq x_i$. Choose $P_i \in \Gamma$ in column x_i as the lowest point $\succeq Q_i$.

Points of $\mathcal{L}$ lost outside $\text{Box}(P_{i-1}, P_i)$ must lie in the P_i^{SE} region bounded vertically between Q_i and P_i . By the value net property, this gap contains at most $\alpha \cdot w(\mathcal{B})/r$ points. Summing over r columns yields $\alpha \cdot w(\mathcal{B})$. □

Structure of the Algorithm

The algorithm is deeply recursive. We evaluate **ApproxLIS**_{*t*}($\mathcal{B}$) to approximate $\text{lis}(\mathcal{B})$ at recursion depth t .

Core procedures:

- 1 **ApproxLIS**_{*t*}: Randomly samples indices to estimate the size of the set $\text{Good}_t(\mathcal{B})$.
- 2 **Classify**_{*t*}($x, \mathcal{B}$): Determines if x is *Good* or *Bad*. Finds a **CriticalBox**_{*t*}(x), then calls **Classify**_{*t-1*}($x, \mathcal{C}$).
- 3 **CriticalBox**_{*t*}: Uses **TerminalBox**_{*t*} and **GridChain**_{*t*}.
- 4 **TerminalBox**_{*t*}: Iteratively splits $\mathcal{B}$ using **FindSplitter** until failure, yielding a Terminal Chain.
- 5 **GridChain**_{*t*}: Builds an α -fine grid in a terminal box, runs **ApproxLIS**_{*t-1*} on grid boxes, and extracts the longest path.

Defining the Approximation Goal

Definition $((\tau, \delta)$ -approximation)

We say $\text{alis}_t(\mathcal{B})$ is a (τ_t, δ_t) -approximation if:

$$|\text{alis}_t(\mathcal{B}) - \text{lis}(\mathcal{B})| \leq \tau_t \text{loss}(\mathcal{B}) + \delta_t w(\mathcal{B})$$

- **Primary Error:** $\tau_t \text{loss}(\mathcal{B})$. τ_t must shrink as t increases.
- **Secondary Error:** $\delta_t w(\mathcal{B})$. Accounts for sampling failure, grid alignment, and splitter violations.

Target: $\tau_t = 4/t$ and $\delta_t = t/\log n$.

Transforming the Goal

Let $\nu_t(\mathcal{B}) = \text{lis}(\mathcal{B}) - \text{alis}_t(\mathcal{B})$. Because $\text{lis}(\mathcal{B}) \geq |\text{Good}_t(\mathcal{B})|$, we define a discrepancy term Δ_t :

$$\Delta_t = \text{lis}(\mathcal{B}) - |\text{Good}_t(\mathcal{B})| - \tau_t \text{loss}(\mathcal{B})$$

Our goal reduces to showing $\Delta_t \leq \delta_t w(\mathcal{B}) - |\epsilon_1|$ (where ϵ_1 is sampling error).

We partition Δ_t over all terminal boxes $\mathcal{T}$ in the terminal chain $\vec{\mathcal{T}}$:

$$\Delta_t(\mathcal{T}) = (1 + \tau_t)|\mathcal{L}(\mathcal{T})| - |\text{Good}_t(\mathcal{B}) \cap \mathcal{T}| - \tau_t \beta(\mathcal{T})$$

where $\mathcal{L}$ is the optimal LIS, and $\beta(\mathcal{T})$ is the number of $\mathcal{F}$ -points in the vertical strip of $\mathcal{T}$.

The Five Secondary Error Terms

To manage the inequality algebraically, we isolate five error components for each terminal box:

- $\epsilon_1 = \text{alis}_t - |\text{Good}_t|$ (Top-level sampling error)
- $\epsilon_2(\mathcal{T}) = \sum_{\mathcal{C}} \text{alis}_{t-1}(\mathcal{C}) - |\text{Good}_{t-1}(\mathcal{C})|$ (Recursive sampling error)
- $\epsilon_3(\mathcal{T}) = |\mathcal{L}^{\text{in}}(\mathcal{T})| - \sum_{\mathcal{E}} |\mathcal{E} \cap \mathcal{L}|$ (Grid alignment loss)
- $\epsilon_4(\mathcal{T}) = |\mathcal{L}^{\text{out}}(\mathcal{T})| - \mu_t \cdot \text{out}(\vec{\mathcal{E}}(\mathcal{T}))$ (Splitter violation loss)
- $\epsilon_5(\mathcal{T}) = \mu_t \cdot \text{aloss}_{t-1}(\vec{\mathcal{E}}) - (1 - \mu_t) \cdot \text{out}(\vec{\mathcal{E}})$ (Dichotomy lemma bound error)

Bounding the Primary Error (Claim 7.3)

By substituting the error terms into $\Delta_t(\mathcal{T})$, we obtain:

$$\begin{aligned}\Delta_t(\mathcal{T}) \leq & (1 + \tau_t)|\mathcal{L}^{out}| - \tau_t \cdot \text{outside} - (1 + \tau_t)\text{loss}(\vec{\mathcal{E}}) \\ & + \text{a loss}_{t-1}(\vec{\mathcal{E}}) + \epsilon_2 + (1 + \tau_t)\epsilon_3\end{aligned}$$

By the inductive hypothesis:

$$\text{loss}(\vec{\mathcal{E}}) \geq \frac{\text{a loss}_{t-1}}{1 + \tau_{t-1}} - \frac{\delta_{t-1}W}{1 + \tau_{t-1}}$$

Let $K_t = 1/(1 + \tau_{t-1})$. Substituting this bounds the a loss term. We then use ϵ_4, ϵ_5 to eliminate $|\mathcal{L}^{out}|$ and a loss_{t-1} .

Setting the τ and μ parameters

The algebraic manipulation yields a coefficient for the $\text{outside}(\vec{\mathcal{E}}(\mathcal{T}))$ term:

$$(1 + \tau_t)\mu_t - \tau_t + [1 - K_t(1 + \tau_t)](\mu_t^{-1} - 1)$$

We MUST make this coefficient ≤ 0 .

- We choose $\mu_t = (1 - K_t)/2$ to minimize the quadratic form.
- This demands the condition:

$$1 + \tau_t \geq \frac{4}{(1 + K_t)^2} \implies \tau_t \geq \left(\frac{2\tau_{t-1} + 2}{\tau_{t-1} + 2} \right)^2 - 1$$

- Guessing $\tau_t = c/t$, we verify that $\tau_t = 4/t$ satisfies the recurrence!
- Consequently, $\mu_t = 2/(t + 3)$.

Bounding Secondary Error: ϵ_4 (Lemma 7.7)

Lemma 7.7

$$\sum_{\mathcal{T}} \epsilon_4(\mathcal{T}) \leq 2\alpha w(\mathcal{B}) + 4 \sum_{P \in \mathcal{P}^\circ} \gamma(P) w(\mathcal{H}(P))$$

Proof Idea.

ϵ_4 bounds points of the LIS falling outside the terminal box but inside its vertical strip. These points form sets $\mathcal{L}^-$ and $\mathcal{L}^+$ bounding the top and bottom.

Because the splitters defining the terminal box were $(\mu, 2\gamma)$ -safe, the number of LIS points violating the corner splitters is strictly bounded by $\gamma \cdot w(\mathcal{H}(P))$. Summing over the depth of the splitter tree yields the $O(\gamma \log n / \rho) w(\mathcal{B})$ bound. □

Bounding Secondary Error: ϵ_5 (Claim 7.8)

Claim 7.8

If $\mathcal{T}$ is wide, $\epsilon_5(\mathcal{T}) \leq 4\rho \cdot w(\mathcal{T})$.

Proof.

We apply the Dichotomy Lemma to the strip $\mathcal{T}|\mathcal{B}$ with grid decomposition Γ .

$$(1 - \mu_t) \cdot \text{out}(\vec{D}) \geq \mu_t |U|$$

Thus, $\epsilon_5(\mathcal{T}) \leq \mu_t(\text{aloss}_{t-1} - |U|) \leq \mu_t |S|$.

Since **FindSplitter** FAILED on $\mathcal{T}$, the number of safe points $|S|$ is bounded by the number of unbalanced points +1, which is $\leq 3\rho w(\mathcal{T}) + 1 \leq 4\rho w(\mathcal{T})$. □

The Bottleneck in the Basic Algorithm

- The running time is $O(\log n)^{O(1/\tau)}$.
- To drop the exponent of $\log n$ to a constant, we must reduce the recursion branching factor.
- The branching factor depends on the grid size $(1/\alpha)^{O(1)}$.
- In the basic algorithm, we needed $\alpha \leq \gamma \approx 1/\log n$. Why? Because applying the dichotomy lemma to bound ϵ_5 required grid boxes to be thinner than the splitter allowance γ .
- If we increase γ , the splitter tree error $\epsilon_4 = \gamma \log n/\rho$ blows up!

The Innovation: Dynamic Phases in **TerminalBox**

Instead of terminating immediately when **FindSplitter** fails, we introduce **Phases**.

- **Phase j :** We allow violation fraction γ_j and balance ρ_j .
- If **FindSplitter** fails, we DO NOT terminate.
- We increment j , exponentially increase $\gamma_j \rightarrow 16\gamma_{j-1}$, and $\rho_j = \gamma_j^{1/4}$.
- We also raise the terminal width threshold θ .
- We terminate only when $w(\mathcal{T}) \leq \theta$.

Consequence: We decouple α from $\log n$. We can set $\alpha = \text{poly}(1/\Psi)$, independent of n .

The Improved **TerminalBox**_t

```
1:  $j \leftarrow 0, \theta \leftarrow \omega, \gamma \leftarrow \gamma_0, \rho \leftarrow \rho_0$ 
2: while  $w(\mathcal{T}) > \theta$  do
3:   Run FindSplitter( $\mathcal{T}, \dots, \gamma, \rho$ )
4:   if splitter found then
5:     Split  $\mathcal{T}$  and update boundaries.
6:   else
7:     // Phase fails. Relax parameters.
8:      $\theta \leftarrow \max(\theta, \gamma w(\mathcal{T})/\alpha)$ 
9:      $j \leftarrow j + 1, \gamma \leftarrow \gamma_j, \rho \leftarrow \rho_j$ 
10:  end if
11: end while
```

Revisiting Tainted Boxes (Lemma 9.7)

Because we shrunk the sample size σ to be independent of n , we can no longer use a union bound over ALL boxes.

Definition (Tainted Box-Level Pair)

$(\mathcal{B}, t)$ is tainted if: 1. Top-level sample fails: $|\text{alis}_t - |\text{Good}_t|| > \Psi_w(\mathcal{B})$. OR 2. Recursive failure: A grid chain exists where the width of tainted children $> \Psi_w(\mathcal{B})$.

Lemma (Lemma 9.7)

The probability that $(\mathcal{B}, t)$ is tainted is $\leq \phi = \alpha^5 / C_2$.

Proof Sketch: Proved via induction on t . Since α -fine grids are disjoint, child failures are independent. A Chernoff bound on independent Bernoulli variables bounds the cascaded failure probability strictly below ϕ .

Fixing the ϵ_5 Bound with Phases

In the basic algorithm, we bounded $|S|$ because **FindSplitter** failed on $\mathcal{T}$. In the improved algorithm, $\mathcal{T}$ terminated because its width dropped below θ , not necessarily because a split failed.

Proposition (Proposition 9.9)

*For any wide terminal box $\mathcal{T}$, there is a **multiphase box** $\mathcal{H}$ (an ancestor in the splitter tree) that failed **FindSplitter** at phase $j - 1$, such that every safe point in $\mathcal{T}$ is (μ_t, γ_{j-1}) -safe for $\mathcal{H}$.*

This allows us to aggregate the ϵ_5 error over the multi-phase boxes $\mathcal{H}$, bounding the total safe points by a geometric series $\sum_j 3\rho_{j-1}w(\mathcal{B})$, which converges to $O(\alpha^{1/4}w(\mathcal{B}))$, completing the final error bound!

Final Time Complexity

- $\Psi = \max(C_2, t_{\max}/\delta)$.
- Grid size is $O(1/\alpha) = O(\Psi^4)$.
- Branching factor per recursive call is purely polynomial in Ψ .
- Recursion depth $t_{\max} = O(1/\tau)$.
- **Total Time:** $(1/\Psi)^{O(1/\tau)} \log^c n = (1/\delta\tau)^{O(1/\tau)} \log^c n$.

Conclusion: A fully sublinear, additive δn approximation to the LIS length!

Questions?

Thank you for your attention.